

Social Media Addiction Risk Predictor

Domain

Healthcare + Behavioral Analytics | Artificial Intelligence | MLOps

Problem Description

In today's digital world, social media platforms have become an inseparable part of daily life, especially among students and young adults. Excessive usage of applications such as Instagram, YouTube, Snapchat, and TikTok can negatively affect mental health, sleep quality, productivity, academic performance, and social interaction.

Studies indicate that prolonged screen time and continuous engagement with social media may lead to addiction-like behavioral patterns including anxiety, stress, lack of concentration, sleep disorders, and emotional instability.

Despite growing awareness, most existing solutions only provide basic screen-time tracking and fail to intelligently analyze behavioral patterns or predict addiction risk levels in real time.

As a result, early warning signs of social media addiction often remain unnoticed until they significantly impact an individual's mental well-being, academic performance, and lifestyle.

Objective

The objective of this project is to design and develop an AI-powered Social Media Addiction Risk Prediction System that analyzes user behavioral patterns and predicts addiction risk levels using machine learning techniques.

The system aims to:

- Monitor user social media usage behavior continuously
- Analyze behavioral and lifestyle patterns intelligently
- Predict addiction risk levels proactively
- Generate a unified addiction risk score
- Provide personalized recommendations for healthy digital habits
- Enable continuous model improvement using an end-to-end MLOps pipeline

Proposed Solution

The proposed system leverages multiple behavioral and lifestyle inputs including:

- Screen Time Data: Daily usage duration, app usage frequency, night-time usage
- Behavioral Data: Mood changes, stress levels, attention span, study interruptions
- Sleep & Lifestyle Data: Sleep duration, productivity level, break frequency
- App Interaction Data: Reels/videos watch time, notification checking frequency, number of app opens
- Survey/Text Inputs: User feedback, emotional patterns, self-reported habits

These inputs are processed using AI and machine learning models to identify addiction-related behavioral patterns and generate a comprehensive addiction risk score (0–1 scale).

System Workflow

1. Data Collection: Collect behavioral and usage-related data from users through forms, surveys, or app tracking systems.
2. Data Preprocessing: Clean, normalize, and transform collected data for model training.
3. Model Processing:
 - Machine Learning models for addiction prediction
 - ANN/RNN models for behavioral pattern analysis
 - Classification models for risk categorization
4. Risk Analysis:
 - Low Risk
 - Moderate Risk
 - High Risk
5. Recommendation Engine: Provide personalized suggestions to improve digital well-being and reduce excessive usage.
6. Dashboard & Visualization: Display analytics including screen-time trends, sleep impact, productivity impact, and weekly addiction score.
7. MLOps Pipeline: Enable continuous monitoring, retraining, deployment, and model versioning.

Expected Outcome

- Predict social media addiction risk accurately
- Help users understand unhealthy digital behavior
- Improve awareness regarding mental wellness and screen-time management
- Provide personalized recommendations for balanced social media usage
- Demonstrate a scalable end-to-end AI and MLOps architecture
- Serve as a foundation for future digital wellness monitoring systems

Innovation & Impact

- Combines behavioral analytics with AI prediction models
- Focuses on mental wellness and digital health awareness
- Implements a complete MLOps lifecycle (data → model → deployment → retraining)
- Provides intelligent personalized recommendations
- Supports scalable deployment for educational institutions and wellness platforms

Technologies Used

AI/ML:

- Python
- Scikit-learn
- TensorFlow / PyTorch

Backend:

- FastAPI

Frontend:

- Streamlit

MLOps Tools:

- MLflow
- Docker
- GitHub Actions

Conclusion

This project addresses the growing concern of social media addiction by proposing an AI-driven behavioral monitoring and prediction system. By leveraging machine learning models and MLOps principles, the system aims to identify addiction risks early, promote healthier digital habits, and improve overall mental well-being among users.